COMPETENCY STANDARD

FOR

CAD CAM DESIGN AND CNC MACHINE OPERATION

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

<u>The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.</u>

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (18 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-CCNC-01-G	Apply Occupational Health and Safety (OHS) Practices in Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	09
SICIP-LE-CCNC-02-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	09
		Total Hour	**18 Hrs.**

Sector Specific Competencies (18 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-CCNC-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	18
		Total Hour	**18 Hrs.**

Occupation Specific Competencies (324 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-CCNC-01-O	Interpret Mechanical Drawings	1. Understand mechanical drawing. 2. Create mechanical drawing.	18
SICIP-LE-CCNC-02-O	Perform CAD Modeling	1. Install and configure CAD software. 2. Use CAD software interface. 3. Create 2D drawings. 4. Develop 3D solid models. 5. Design mold in CAD. 6. Generate 2D drawing sheet. 7. Save and manage CAD files.	90
SICIP-LE-CCNC-03-O	Select Cutting Tools and Workpieces	1. Understand cutting tools, inserts and tool materials. 2. Select cutting tools and parameters for CNC lathe. 3. Select cutting tools and parameters for CNC milling. 4. Select suitable workpiece materials.	18
SICIP-LE-CCNC-04-O	Perform CAM Programming	1. Install and configure CAM software. 2. Use CAM software interface and features. 3. Carry out 2D milling CAM programming. 4. Carry out 3D milling CAM programming. 5. Carry out lathe CAM programming. 6. Generate and simulate CNC programs. 7. Understand G&M Codes.	108
SICIP-LE-CCNC-05-O	Operate CNC Milling Machine	1. Prepare CNC milling machine. 2. Load and transfer CNC programs. 3. Perform CNC milling operations. 4. Check and measure workpiece.	45
SICIP-LE-CCNC-06-O	Operate CNC Lathe Machine	1. Prepare CNC lathe machine.	45

		2. Load and transfer CNC programs. 3. Perform CNC lathe operations. 4. Check and measure workpiece.	
		Total Hour	**324 Hrs.**

COMPETENCY STANDARD: CAD CAM DESIGN AND CNC MACHINE OPERATION

The Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE**	Nominal Duration: 09 hrs.	Unit Code: SICIP-LE-CCNC-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 **<u>Personal Protective Equipment (PPE)</u>** is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 **<u>Hazards and risks</u>** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 **<u>Emergency response plans and procedures</u>** are implemented. 4.3 **<u>First aid procedure</u>** is applied during emergency situations.

Range of Variables

Variable	Range May include but not limited to:

1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.3.1 Machine hazards 3.3.2 Materials hazards 3.3.3 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Explosion response 1.3.4 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects. 1.5 PPE types and uses 1.6 Personal hygiene practices 1.7 OHS awareness 1.8 Malfunctions and resolutions
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices in the workplace 2.3 Identifying, assessing and controlling hazards and risks 2.4 Reporting incidents arising from hazards and risks 2.5 Responding to emergency procedures

	2.6 Maintaining physical well-being in the workplace 2.7 Using firefighting accessories and fire extinguishers 2.8 Implementing emergency response plans and procedures 2.9 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Learning manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described emergency procedures 1.2 followed OHS policies and procedures 1.3 selected and used personal protective equipment (PPE) 1.4 practiced personal health and safety procedures 1.5 identified, assessed and controlled hazards and risks 1.6 reported incidents arising from hazards and risks to the authority 1.7 implemented plans and procedures to respond emergency 1.8 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A SELF-DIRECTED TEAM**	**Nominal Duration:** 09 hrs.	**Unit Code:** SICIP-LE-CCNC-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal **forms of communication** are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range May Include but not limited to:
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OHS information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Methods of identifying current and potential problems faced by a team 1.10 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice

	2.8 Appreciating and valuing team member's support to other members of the team 2.9 Identifying current and potential problems faced by the team 2.10 Identifying solutions to the problem 2.11 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work book 4.3 Learning & operational manuals 4.4 Workplace tools/equipment

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described team's duties, responsibilities, authorities, objectives and task requirements 1.2 identified team goals and work processes 1.3 communicated and cooperated with team members 1.4 used effective interpersonal skills to interact with team members 1.5 worked as a team member 1.6 solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

The Sector Specific Competencies

Unit of Competency: **APPLY GREEN PRACTICES**	**Nominal Duration:** 18 hrs.	**Unit Code:** SICIP-LE-CCNC-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices</u>** are explained. 1.2 **<u>Sources of environmental impacts</u>** during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 **<u>Ways of mitigating environmental impacts</u>** in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy and raw material consumption are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of wastes</u>** are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce wastes in personal and professional life are practiced.

Range of Variables

Variable	**Range** May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle

	1.2.5 Recover 1.2.6 Repair 1.3 Use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical’s leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 Waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment 3.2 Adopting renewable energy sources 3.3 Implementing site protection measures 3.4 Using reusable materials 3.5 Choosing sustainable materials 3.6 Using noise-reducing equipment and scheduling work appropriately 3.7 Optimizing logistics and delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of wastes	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in light engineering sector 1.2 Ways of mitigating environmental impacts in light engineering sector 1.3 Key terms and symbols related to environmental sustainability in light engineering drawings 1.4 Sources of environmental impacts in light engineering 1.5 Methods to minimize resource consumption (water, energy, raw materials) 1.6 Waste management practices 1.7 Sustainable alternatives to fossil-based energy resources 1.8 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying sources of environmental impacts during light engineering activities 2.2 Identifying recyclable and non-recyclable items 2.3 Exploring and applying sustainable alternatives to fossil-based energy resources 2.4 Applying methods to minimize resource use in the workplace 2.5 Implementing waste management practices 2.6 Applying procedures for safely disposing of hazardous materials 2.7 Identifying different types of wastes 2.8 Documenting water, energy and material consumption in the workplace 2.9 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of light engineering hand tools and power tools 4.3 Sustainable, re-cyclable, reusable materials 4.4 Work books 4.5 Operation and maintenance manuals 4.6 Waster segregation bins 4.7 Learning manual

	4.8 Energy-efficient equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 explained the principles of green practices in light engineering 1.2 identified sources of environmental impacts 1.3 implemented procedures to minimize resource use (water, energy, raw materials) 1.4 explained ways of mitigating environmental impacts in light engineering sector 1.5 disposed of hazardous waste in compliance with safety and environmental standards 1.6 practiced green habits to reduce waste in both personal and professional life 1.7 applied methods to minimize resource use in the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **INTERPRET MECHANICAL DRAWINGS**	**Nominal Duration:** 18 hrs.	**Unit Code:** SICIP-LE-CCNC-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret mechanical drawings. It specifically includes the tasks of understanding mechanical drawing and creating mechanical drawing.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand mechanical drawing	1.1 Different types of **<u>mechanical drawings</u>** are interpreted. 1.2 Different **<u>views of mechanical drawing</u>** are identified and described. 1.3 Necessity of putting dimensions in different drawings is comprehended.
2. Create mechanical drawing	2.1 Components, objects, parts, materials and assemblies are identified, drawn and described. 2.2 Base line or datum points are specified, as required. 2.3 Instructions are included in drawing. 2.4 Drawing is created accurately. 2.5 **<u>Tolerance</u>**, limits and fits are identified in technical drawing.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Mechanical drawings	1.1 Sketch components 1.1.1 Line 1.1.2 Rectangle 1.1.3 Circle 1.1.4 Ellipse 1.1.5 Polygon 1.1.6 Arc 1.2 Elements to determine dimension 1.2.1 Central line 1.2.2 Object line 1.2.3 Tolerance 1.2.4 Ladder line 1.2.5 Extension line

	1.2.6 Hidden line 1.2.7 Cutting plane line 1.2.8 Short brake 1.2.9 Long brake
2. Views of mechanical drawing	2.1 Top and Bottom View 2.2 Right and Left View 2.3 Front View 2.4 Sectional view 2.5 Isometric View 2.6 Oblique View 2.7 Hidden view 2.8 Perspective view
3. Tolerance	3.1 Dimensional tolerance 3.2 Geometric tolerance (GD&T)

Curricular Content Guide

1. Underpinning knowledge	1.1 Basic concept of mechanical drawings and their role in manufacturing and machining work. 1.2 Understanding of different types of mechanical drawings. 1.3 Identification of different drawing views. 1.4 Purpose of dimensions in mechanical drawings and how they define size and shape. 1.5 Understanding of symbols, notes, and instructions used in technical drawings. 1.6 Understanding of tolerances, limits, and fits and their importance in part interchangeability. 1.7 Knowledge of how mechanical drawings guide CAD design and CNC machining operations.
2. Underpinning skills	2.1 Interpreting different types of mechanical drawings. 2.2 Identifying and understanding orthographic and sectional views. 2.3 Understanding the purpose and placement of dimensions. 2.4 Drawing components, parts, and assemblies accurately. 2.5 Defining datum points and reference lines correctly. 2.6 Applying tolerances, limits, and fits in technical drawings. 2.7 Including clear instructions and annotations in drawings.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Drawing tools and supplies 4.4 Specimen parts and components 4.5 Drawings and sketches

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 understood mechanical drawing. 1.2 created mechanical drawing.
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM CAD MODELING**	**Nominal Duration:** 90 hrs.	**Unit Code:** SICIP-LE-CCNC-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform CAD modeling. It specifically includes the tasks of installing and configuring CAD software, using CAD software interface, creating 2D drawings, developing 3D solid models, designing mold in CAD, generating 2D drawing sheet and saving and managing CAD files.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install and configure CAD software.	1.1 **Components of computer** are identified and interpreted. 1.2 **CAD software** is installed as per standard operating procedure. 1.3 Screen display areas and **basic parameters** are set as per job specification.
2. Use CAD software interface.	2.1 CAD software is opened and the correct workspace is selected. 2.2 Menus, toolbars, part feature manager and command panels are identified. 2.3 Drawing units and snap settings are set as required. 2.4 **Basic navigation tools** and status bar are identified. 2.5 Software settings are adjusted according to design requirements.
3. Create 2D drawings.	3.1 **Sketch tools** are identified and selected for 2D drawing. 3.2 **Sketch modify tools** are identified and selected for 2D drawing. 3.3 2D **sketch relations** are identified and explained. 3.4 2D drawing is created. 3.5 CAD drawing is reviewed and modified, as necessary.
4. Develop 3D solid models.	4.1 **Features tools** are identified and selected for 3D model. 4.2 **Direct editing tools** are identified and selected for 3D model. 4.3 3D model is developed.
5. Design mold in CAD.	5.1 3D model is reviewed and modified, if necessary. 5.2 2D drawing is generated for review and finalizing the model. 5.3 Mold is created based on the 3D model. 5.4 A separate folder is developed and the mold is saved for CAM programming.
6. Generate 2D drawing sheet.	6.1 Drawing environment is prepared according to standards. 6.2 Standard drawing views are inserted.

	6.3 Section and detail views are applied as required. 6.4 Annotations and standard symbols are added. 6.5 Tolerances and notes are applied according to specifications. 6.6 Title block information is completed accurately.
7. Save and manage CAD files.	7.1 CAD files are saved using correct file names and **file formats.** 7.2 File locations and folders are organized properly. 7.3 Version control is maintained to track design changes. 7.4 Backup copies are created to prevent data loss. 7.5 Files are exported or shared as required.

Range of Variables

Variable	**Range** May include but not limited to:
1. Components of computer	1.1 Hardware 1.1.1 RAM 1.1.2 Processor 1.1.3 Graphic Card 1.1.4 Storage 1.1.5 Monitor
2. CAD software	2.1 SolidWorks 2.2 Siemens NX 2.3 CATIA 2.4 Auto CAD 2.5 Fusion 360
3. Basic parameters	3.1 User interface 3.2 Units (mm or inch) 3.3 Color and text format
4. Basic navigation tools	4.1 Zoom 4.2 Pan 4.3 Rotate
5. Sketch tools	5.1 Lines 5.2 Rectangle 5.3 Arc 5.4 Circle 5.5 Polygon 5.6 Ellipse 5.7 Spline 5.8 Slot
6. Sketch modify tools	6.1 Trim 6.2 Offset 6.3 Mirror 6.4 Linear and circular pattern 6.5 Move, copy, rotate, scale, extend and stretch

7. Sketch relations	7.1 Horizontal 7.2 Vertical 7.3 Coincident 7.4 Intersection 7.5 Tangent 7.6 Collinear 7.7 Parallel 7.8 Concentric 7.9 Pairs 7.10 Symmetric
8. Feature tools	8.1 Extruded boss/base 8.2 Extruded cut 8.3 Revolved boss/base 8.4 Revolved cut 8.5 Swept boss/base 8.6 Swept cut 8.7 Lofted boss/base 8.8 Lofted cut 8.9 Boundary boss/base 8.10 Boundary cut
9. Direct editing tools	9.1 Fillet, Chamfer, Rib, Draft, Shell, Wrap, Mirror, Pattern 9.2 Flex 9.3 Surface tools 9.4 Mold tools 9.5 Move face 9.6 Move/copy bodies 9.7 Delete face 9.8 Split 9.9 Combine
10. File formats	10.1 .SLDPRT 10.2 .SLDASM 10.3 .SLDDRW 10.4 .STEP / .STP 10.5 iges 10.6 Parasolid 10.7 .PDF

Curricular Content Guide

1. Underpinning knowledge	1.1 Basic concept of CAD software and its role in design, modeling, and manufacturing. 1.2 Understanding of computer requirements and system setup. 1.3 Knowledge of CAD software interface elements. 1.4 Concept of drawing units, snap settings, and basic navigation for accurate drafting. 1.5 Understanding of 2D sketching concepts, sketch tools, modify tools, and sketch relations.

	1.6 Basic idea of creating and reviewing accurate 2D drawings in CAD. 1.7 Concept of 3D solid modeling using feature tools and direct editing tools.
2. Underpinning skills	2.1 Installing and configuring CAD software and system settings. 2.2 Navigating and using the CAD software interface effectively. 2.3 Creating and modifying accurate 2D drawings. 2.4 Developing 3D solid models using feature and editing tools. 2.5 Designing molds based on 3D models for CAM preparation. 2.6 Generating standardized 2D drawing sheets with annotations and tolerances. 2.7 Saving, organizing, managing, and backing up CAD files properly.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 CAD software 4.4 Drawings and sketches

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 installed and configured CAD software. 1.2 used CAD software interface. 1.3 created 2D drawings. 1.4 developed 3D solid models. 1.5 designed mold in CAD. 1.6 generated 2D drawing sheet. 1.7 saved and managed CAD files.
2. Methods of assessment	Methods of assessment may include but is not limited to:

	2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SELECT CUTTING TOOLS AND WORKPIECES**	**Nominal Duration:** 18 hrs.	**Unit Code:** SICIP-LE-CCNC-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to select cutting tools ad workpieces. It specifically includes the tasks of understanding cutting tools, insert and tools materials, selecting cutting tools and parameters for CNC lathe and selecting cutting tools & parameters for CNC milling.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand cutting tools, insert and tool materials.	1.1 **<u>Geometry of cutting tools</u>** is identified. 1.2 **<u>Types of cutting tools and inserts</u>** are recognized. 1.3 Tool materials and coatings are identified. 1.4 Tool application is matched with workpiece material.
2. Select cutting tools and parameters for CNC lathe.	2.1 **<u>Cutting tools for lathe operations</u>** are identified. 2.2 **<u>Workpiece material</u>** and specifications are identified. 2.3 Appropriate cutting tools are selected based on material and operation. 2.4 **<u>Cutting parameters</u>** are determined.
3. Select cutting tools and parameters for CNC milling.	3.1 **<u>Cutting tools for CNC milling operations</u>** are identified. 3.2 Workpiece material and machining requirements are identified. 3.3 Appropriate milling tools are selected based on operation type. 3.4 Cutting parameters are determined.

Range of Variables

Variable	Range May include but not limited to:
1. Geometry of cutting tools	1.1 Number of flutes 1.2 Diameter and length 1.3 Material of cutting tool 1.3.1 High speed steel 1.3.2 Carbide 1.4 Angles of cutting tool 1.4.1 Rack angle 1.4.2 Clearance angle 1.4.3 Relief angle
2. Types of cutting tools and inserts	2.1 Single point 2.1.1 Turning 2.1.2 Facing

	2.1.3 Boring 2.1.4 Threading 2.2 Multi point 2.2.1 Solid end mill 2.2.2 Ball mill 2.2.3 Radius mill 2.2.4 Face mill 2.2.5 Drill bit 2.2.6 Reamer 2.2.7 Tap and die 2.3 Type of insert 2.3.1 CNMG 2.3.2 TNMG 2.3.3 DNMG 2.3.4 VNMG
3. Cutting tools for lathe operations	3.1 Turning tools 3.2 Facing tools 3.3 Grooving tools 3.4 Threading tools 3.5 Boring tools
4. Workpiece material	4.1 Aluminum 4.2 Mild steel 4.3 Teflon
5. Cutting parameters	5.1 Cutting speed 5.2 Feed 5.3 Depth of cut 5.4 RPM
6. Cutting tools for CNC milling operations	6.1 Face mill 6.2 End mill 6.3 Bull mill 6.4 Ball mill 6.5 Machine tap

Curricular Content Guide

1. Underpinning knowledge	1.1 Basic concept of cutting tools, inserts, and their role in CNC machining. 1.2 Knowledge of common tool materials and coatings and their performance characteristics. 1.3 Concept of matching cutting tools with workpiece materials for effective machining. 1.4 Understanding of CNC lathe cutting tools and their application in turning operations. 1.5 Basic idea of CNC milling cutting tools and their use in different milling operations. 1.6 Concept of cutting parameters.
2. Underpinning skills	2.1 Identifying cutting tool geometry, types, and inserts.

	2.2 Recognizing tool materials and surface coatings. 2.3 Matching cutting tools with workpiece materials. 2.4 Selecting appropriate cutting tools for CNC lathe operations. 2.5 Determining cutting parameters for CNC lathe machining. 2.6 Selecting suitable cutting tools for CNC milling operations. 2.7 Determining cutting parameters for CNC milling processes.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Learning manual 4.4 Software, hardware 4.5 System compatibility, protocols 4.6 Drawings, Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 understood cutting tools, insert and tool materials. 1.2 selected cutting tools and parameters for CNC lathe. 1.3 selected cutting tools and parameters for CNC milling.
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM CAM PROGRAMMING**	**Nominal Duration:** 108 hrs.	**Unit Code:** SICIP-LE-CCNC-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform CAM programming. It specifically includes the tasks of installing and configuring CAM software, using CAM software interface and features, carrying out 2D milling CAM programming, carrying out 3D milling CAM programming, carrying out lathe CAM programming, generating and simulating CNC programs and understanding G&M codes.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install and configure CAM software.	1.1 Materials, instructions and standard operating procedure are obtained according to job requirement. 1.2 CAM package is installed as per standard operating procedure. 1.3 **<u>System parameters</u>** are identified and selected according to job requirement.
2. Use CAM software interface and features.	2.1 **<u>Software interface components</u>** are identified and described. 2.2 CAM software interface is used as per job requirements.
3. Carry out 2D milling CAM programming.	3.1 CAD model is imported and oriented. 3.2 Reference point is established or originated based on job requirement. 3.3 Stock set-up is performed. 3.4 Cutting tools are identified and selected. 3.5 Sequential **<u>2D toolpaths for milling operations</u>** are identified, generated and verified. 3.6 NC codes are generated.
4. Carry out 3D milling CAM programming.	4.1 3D CAD model is imported and oriented. 4.2 Reference point is established or originated based on job requirement. 4.3 Stock set-up is performed. 4.4 Cutting tools are identified and selected 4.5 Sequential **<u>3D toolpaths for milling operations</u>** are identified, generated and verified. 4.6 G code and M code are generated.
5. Carry out lathe CAM programming.	5.1 CAD model is imported and oriented. 5.2 Reference point is established or originated based on job requirement. 5.3 Stock set-up is performed.

	5.4 Cutting tools are identified and selected. 5.5 Sequential **2D toolpaths for lathe operations** are identified, generated and verified. 5.6 NC codes are generated.
6. Generate and simulate CNC programs.	6.1 Program is loaded using appropriate **device**. 6.2 Dry run/simulation is performed in computer and machine as per standard operating procedure. 6.3 Program is executed to produce work piece. 6.4 **Problems** encountered are recorded and reported to appropriate authority as per standard operating procedure.
7. Understand G&M Codes.	7.1 **G codes** are identified and interpreted. 7.2 **M codes** are identified and interpreted. 7.3 Functions of G codes in motion and control are understood. 7.4 Functions of M codes for machine operation are understood.

Range of Variables

Variable	**Range** May include but not limited to:
1. System parameters	1.1 Metric or English system 1.2 Tool bars (dimensioning, line types, editing, hatching)
2. Software interface components	2.1 Wireframe 2.1.1 Line 2.1.2 Rectangle 2.1.3 Circle 2.1.4 Polygon 2.1.5 Slot 2.1.6 Arc 2.1.7 Rectangular shape 2.1.8 Carve on one edge 2.1.9 Bounding box 2.1.10 Turn profile 2.2 Surface 2.2.1 Move face 2.2.2 Delete face 2.2.3 Push/pull 2.2.4 Modify 2.3 Solid 2.3.1 Extrude boss/cut 2.3.2 Revolve boss/cut 2.3.3 Lofted boss/cut 2.4 Transform 2.4.1 Translate 2D 2.4.2 Translate 3D

	2.4.3 Dynamic transform 2.4.4 Move to origin 2.4.5 Offset 2.4.6 Mirror 2.4.7 Pattern 2.5 Machine 2.5.1 CNC default lathe machine 2.5.2 CNC default milling machine 2.6 Toolpath 2.6.1 Facing 2.6.2 Pocket milling 2.6.3 Contour 2.6.4 Parallel 2.6.5 Scallop 2.6.6 Drilling
3. 2D toolpaths for milling operations	3.1 Facing 3.2 Pocket milling 3.2.1 Circular milling 3.2.2 Slot milling 3.3 Contour 3.4 Drilling
4. 3D toolpaths for milling operations	4.1 Roughing 4.1.1 Surface rough pocket 4.1.2 Surface rough parallel 4.2 Finishing 4.2.1 Surface finish contour 4.3 Surface finish scallop
5. 2D toolpaths for lathe operations	5.1 Roughing 5.1.1 Turning 5.1.2 Boring 5.2 Finishing 5.2.1 Turning 5.2.2 Boring 5.3 Drilling 5.3.1 Direct 5.3.2 Peck 5.4 Threading 5.4.1 External 5.4.2 Internal 5.5 Grooving 5.6 Parting
6. Device	6.1 USB drive 6.2 Memory card 6.3 Portable hard disk 6.4 Networking
7. Problems	7.1 Incorrect machine set-up 7.2 Incorrect parameter setting

	7.3 Incorrect tool and work offset 7.4 Defective raw materials
8. G codes	8.1 G00 – Rapid Positioning 8.2 G01 – Linear Interpolation 8.3 G02 – Circular Interpolation (Clockwise) 8.4 G03 – Circular Interpolation (Counter-Clockwise) 8.5 G17 – XY Plane Selection 8.6 G18 – XZ Plane Selection 8.7 G19 – YZ Plane Selection 8.8 G20 – Inch Programming 8.9 G21 – Millimeter Programming 8.10 G28 – Return to Machine Zero 8.11 G40 – Cancel Cutter Radius Compensation 8.12 G41 – Cutter Radius Compensation Left 8.13 G42 – Cutter Radius Compensation Right 8.14 G43 – Tool Length Compensation On 8.15 G49 – Tool Length Compensation Cancel 8.16 G54 – Work Coordinate System 1 8.17 G55 – Work Coordinate System 2 8.18 G56 – Work Coordinate System 3 8.19 G57 – Work Coordinate System 4 8.20 G58 – Work Coordinate System 5 8.21 G59 – Work Coordinate System 6 8.22 G90 – Absolute Positioning 8.23 G91 – Incremental Positioning
9. M codes	9.1 M01 – Optional Program Stop 9.2 M02 – End of Program 9.3 M03 – Spindle On (Clockwise) 9.4 M04 – Spindle On (Counter-Clockwise) 9.5 M05 – Spindle Stop 9.6 M06 – Tool Change 9.7 M08 – Coolant On 9.8 M09 – Coolant Off 9.9 M30 – End of Program and Reset

Curricular Content Guide

1. Underpinning knowledge	1.1 Basic concept of CAM software and its role in converting CAD designs into CNC machining programs. 1.2 Installation procedures, and configuration parameters for CAM software. 1.3 Knowledge of CAM software interface elements, features. 1.4 Concept of importing CAD models, setting orientation, reference points, and stock setup in CAM. 1.5 Understanding of cutting tool selection and its relationship with machining operations in milling and turning. 1.6 Basic idea of 2D milling, 3D milling, and lathe CAM

	programming workflows. 1.7 Concept of toolpaths, sequencing of operations, and verification to ensure correct machining. 1.8 Understanding of NC program generation, simulation, dry run, and error detection before machining. 1.9 Knowledge of G-codes and M-codes, their functions in machine motion, control, and machine operations.
2. Underpinning skills	2.1 Installing and configuring CAM software according to job requirements. 2.2 Navigating and using CAM software interface and features. 2.3 Creating and verifying 2D milling CAM programs and toolpaths. 2.4 Creating and verifying 3D milling CAM programs and toolpaths. 2.5 Developing CAM programs for CNC lathe operations. 2.6 Generating, simulating, and executing CNC programs safely. 2.7 Interpreting and applying G and M codes for CNC machining operations.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Drawings, job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 installed and configured CAM software. 1.2 used CAM software interface and features. 1.3 carried out 2D milling CAM programming. 1.4 carried out 3D milling CAM programming. 1.5 carried out lathe CAM programming. 1.6 generated and simulated CNC programs. 1.7 understood G&M Codes.

2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE CNC MILLING MACHINE**	**Nominal Duration:** 45 hrs.	**Unit Code:** SICIP-LE-CCNC-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to operate CNC milling machine. It specifically includes the tasks of preparing CNC milling machine, loading and transferring CNC programs, performing CNC milling operations and checking and measuring workpiece.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC milling machine.	1.1 Main switch and emergency switch are turned on and power supply is checked. 1.2 Lubrication, air, hydraulic pressure, and coolant levels are checked as per specifications. 1.3 Machine zero point is checked and confirmed. 1.4 **CNC machine controller** interface is identified and basic functions are verified. 1.5 **Cutting tools** are set and loaded according to the required sequence of operation. 1.6 **Clamping devices** are installed and tightened following standard procedures. 1.7 Workpiece is mounted, aligned, and part and tool zero settings are performed accurately.
2. Load and transfer CNC programs.	2.1 Program file is loaded and transferred to the CNC machine using appropriate device. 2.2 Program name, version, and format are verified. 2.3 Program is checked for errors or missing data. 2.4 Missing data is verified and edited if necessary. 2.5 Correct program is confirmed before execution.
3. Perform CNC milling operations.	3.1 **CNC milling operations** are performed according to the loaded program. 3.2 Machine operation is monitored during machining. 3.3 Corrections are made when deviations are observed. 3.4 Cutting conditions are adjusted if required. 3.5 Machined components are produced as per drawing and specifications.
4. Check and measure workpiece.	4.1 Work piece is checked and measured against specification using appropriate methods and measuring tools. 4.2 Defective work pieces are marked, recorded and reported for proper action.

Range of Variables

Variable	Range May include but not limited to:
1. CNC machine controller	1.1 Haas 1.2 Fanuc 1.3 Siemens 1.4 Mitsubishi 1.5 GSK
2. Cutting tools	2.1 Face/Bull Mill 2.2 End Mill 2.3 Ball mill 2.4 Tapper mill 2.5 Drill Bits 2.6 Reamers
3. Clamping devices	3.1 Angle plate 3.2 V-Block, U-Clamp and C-Clamp 3.3 Step-Block 3.4 T-Slot bolt 3.5 Machine vice 3.6 Pneumatic fastening clamps 3.7 Jig and fixtures
4. CNC milling operations	4.1 Facing 4.2 Pocket 4.3 Drilling 4.4 Contour 4.5 3D toolpaths

Curricular Content Guide

1. Underpinning knowledge	1.1 Basic concept of CNC milling machines and their role in precision machining. 1.2 Understanding of machine power systems, lubrication, coolant, air, and hydraulic requirements. 1.3 Knowledge of machine zero, work zero, and tool zero and why they are critical for accuracy. 1.4 Understanding of CNC controller interface, basic controls, and program execution functions. 1.5 Concept of cutting tool selection, tool sequencing, and tool holding systems. 1.6 Knowledge of clamping devices, work holding methods, and proper alignment of workpieces. 1.7 Understanding of CNC program loading, verification, version control, and error checking. 1.8 Concept of CNC milling operations, machine monitoring, and adjustment during machining. 1.9 Knowledge of workpiece inspection, measurement methods, and identification of defects against drawings

	and specifications.
2. Underpinning skills	2.1 Preparing and setting up the CNC milling machine safely. 2.2 Checking lubrication, coolant, air, and hydraulic systems. 2.3 Setting cutting tools, clamping devices, and workpiece alignment. 2.4 Loading, transferring, and verifying CNC programs. 2.5 Operating CNC milling machines according to programs. 2.6 Monitoring machining processes and making necessary adjustments. 2.7 Measuring finished workpieces and identifying defects.
3 Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4 Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 CNC milling machine 4.4 Drawings, specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 prepared CNC milling machine. 1.2 loaded and transferred CNC programs. 1.3 performed CNC milling operations. 1.4 checked and measured workpiece.
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE CNC LATHE MACHINE**	**Nominal Duration:** 45 hrs.	**Unit Code:** SICIP-LE-CCNC-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to operate CNC lathe machine. It specifically includes the tasks of preparing CNC lathe machine, loading and transfer CNC programs, performing CNC lathe operations and checking and measuring workpiece.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC lathe machine.	1.1 Main switch and emergency switch are turned on and power supply is checked. 1.2 Lubrication, air, hydraulic pressure, and coolant levels are checked as per specifications. 1.3 Machine zero point is checked and confirmed. 1.4 CNC controller interface is identified and basic functions are verified. 1.5 **<u>Cutting tools</u>** are set according to the required sequence of operation. 1.6 **<u>Clamping devices</u>** are installed and tightened following standard procedures. 1.7 Workpiece is mounted, centered, and part zero setting is performed accurately.
2. Load and transfer CNC programs.	2.1 CNC program is selected according to job requirements. 2.2 Program file is transferred to the CNC lathe using an approved method. 2.3 Program is loaded into the controller memory correctly. 2.4 Program name, format, and version are verified. 2.5 Program is checked for errors before execution. 2.6 Missing data is verified and edited if necessary. 2.7 Correct program is confirmed prior to machining.
3. Perform CNC lathe operations.	3.1 **<u>CNC lathe operations</u>** are performed according to the loaded program. 3.2 Components are machined as per drawings and specifications. 3.3 Machine operation is monitored during machining. 3.4 Cutting conditions are adjusted when required. 3.5 **<u>Corrective measures</u>** are taken if deviations are observed.

4. Check and measure workpiece.	4.1 Work piece is checked and measured against specification using appropriate methods and measuring tools. 4.2 Defective work pieces are marked, recorded and reported for proper action.

Range of Variables

Variable	**Range** May include but not limited to:
1. Cutting tools	1.1 Facing 1.2 Turning 1.3 Grooving 1.4 Drilling 1.5 Internal Threading 1.6 External Threading 1.7 Parting-off 1.8 Boring 1.9 Knurling 1.10 Finishing
2. Clamping devices	2.1 Three jaw chuck 2.2 Four jaw chuck 2.3 Collet chuck 2.4 Live center 2.5 Bar feeder 2.6 Part catcher 2.7 Tool center
3. CNC lathe operations	3.1 Facing 3.2 Turning 3.3 Tapper Turning 3.4 Threading 3.5 Drilling 3.6 Boring 3.7 Grooving 3.8 Parting-off
4. Corrective measures	4.1 Cutting tool changes 4.2 Adjustment of tool offset 4.3 Adjustment of cutting speed and feed rate

Curricular Content Guide

1. Underpinning knowledge	1.1 Basic concept of CNC lathe machines and their use in cylindrical machining operations. 1.2 Understanding of machine power supply, lubrication, hydraulic, air, and coolant systems. 1.3 Knowledge of machine zero, part zero, and their role in machining accuracy.

	1.4 Understanding of CNC controller interface, basic controls, and operational functions. 1.5 Concept of cutting tool types, tool sequence, and tool holding systems used in CNC lathe work. 1.6 Knowledge of work holding methods, clamping devices, centering, and alignment of workpieces. 1.7 Understanding of CNC program selection, loading, verification, and version control. 1.8 Concept of CNC lathe operation, monitoring machining processes, and adjusting cutting conditions.
2. Underpinning skills	2.1 Preparing and setting up the CNC lathe machine safely. 2.2 Checking lubrication, coolant, air, and hydraulic systems. 2.3 Setting cutting tools, clamping devices, and workpiece alignment. 2.4 Loading, transferring, and verifying CNC lathe programs. 2.5 Operating the CNC lathe according to program instructions. 2.6 Monitoring machining processes and adjusting cutting conditions. 2.7 Measuring finished components and identifying defects.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Learning manual 4.4 CNC lathe machine

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 prepared CNC lathe machine. 1.2 loaded and transferred CNC programs. 1.3 performed CNC lathe operations. 1.4 checked and measured workpiece.
2. Methods of assessment	Methods of assessment may include but is not limited to:

	2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	CAD CAM Design and CNC Machine Operation
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Workstation Computer with CAD and CAM Software	26
2.	CNC Lathe Machine	1
3.	CNC Milling Machine	1
4.	Bench Vice	5
5.	Working Table	2
6.	Cutting tools set for lathe machine	2
7.	Cutting tools set for milling machine	2
8.	Vernier caliper	5
9.	Micrometer	5
10.	Tool Grinding Machine	1
11.	Clamping Device	2 Set
12.	Machine vice	1
13.	Hand grinding machine	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on CAD CAM Design and CNC Machine Operation, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.